Mini-Research 8

Physical Feasibility, Extreme Temperature Limits, and Energy Constraints

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Abstract

I examine which temperature statements remain defensible in a hierarchy of acoustic bubble and laser–gold–liquid–gas models. The bounded new calculations are a one-cycle Keller–Miksis/Gilmore–Tait comparison at 1.00, 1.15 and 1.30 bar with a common polytropic gas; signed energy and nonlinear caloric audits of saved coupled trajectories; and particle-retention and transport-scale diagnostics. At 1.30 bar Gilmore reduces the minimum radius by about 6.80% relative to Keller–Miksis while changing pressure and local sound speed substantially. No temperature is transplanted between trajectories. For the nominal 34.3373 fJ single-particle absorption, ideal instantaneous frozen-gas heating at fixed volume gives 0.03347 and 0.02499 K at the two principal wet reference states. The actual integrated global changes remain 1.408816 and 2.184879 microkelvin; the latter is physically formal despite numerical completion. Wet 1.30 bar censoring at 6000 K is a coefficient boundary, not a physical ceiling. Formal dry and incompressible extremes are preserved. No universal physical upper temperature, local ion state or thermonuclear yield can be established from the available information.

1 Physical question and scope

The program asks whether a laser, mediated by Au nanoparticles, can modify acoustic compression sufficiently to approach thermonuclear conditions. The pathway is absorption, Au storage, water transport, gas-boundary transfer and mechanical feedback. Available optical energy is not synonymous with gas heat or an increased global temperature maximum.

The inherited reference is $R_0 = 6\,\mu$m, $f = 30$ kHz, $T_0 = 293.15$ K and $p_\infty = p_0 - P_a \sin(2\pi f t)$, with $p_0 = 101325$ Pa. The nominal optical pair uses $a = 20$ nm, prescribed 100 nm surface gap, $\lambda_0 = 532$ nm, a 10 ns top hat and $F = 10$ J/m^2. Mini-Research 7 gives complete five-cycle global increments $+1.408816$ and $+2.184879\,\mu$K at 1.00 and 1.15 bar; wet 1.30 bar has no global increment after coefficient censoring.[1] These results are retained without refitting. The present work does not implement new CFD, a reacting-plasma model or a fusion simulator. New mechanics uses one complete acoustic cycle and the common earlier polytropic gas to isolate liquid-model sensitivity. All other new numerical analysis uses saved public traces and coefficients.

Four interpretation categories are used: validated within the declared mathematical model; physically grounded conditional prediction; numerically formal extrapolation or extremum; and unresolved physical scenario. Numerical and physical categories are reported separately. None of them implies experimental validation.

[1] Mini-Research 7, "Integrated Laser–AuNP–Bubble Model: Self-Consistent Dynamics and Peak Gas Temperature," Section "Paired temperatures and timing sensitivity," `07_Integrated_Model.pdf`.

2 Extreme mechanics and a controlled Gilmore comparison

2.1 Liquid closure and common gas

The comparison uses unchanged mechanical water constants $\rho_0 = 998.2$ kg/m^3, $c_0 = 1482$ m/s, $\mu = 0.001002$ Pa s and $\sigma = 0.07274$ N/m. Both models start at $R_0, U = 0$ and use

$$p_d = p_{d0}(R_0/R)^{3\kappa}, \quad p_{d0} = p_0 + 2\sigma/R_0 - p_v, \quad \kappa = 1.4, \quad p_v = 2339 \text{ Pa}. \tag{1}$$

This closure is from the earlier mechanical comparison, not the later wet-mixture gas-energy model.[2] Define $p_L = p_d + p_v - 2\sigma/R - 4\mu U/R$ and $\Delta p = p_L - p_\infty$. KM is

$$(1 - U/c_0)R\dot{U} + \tfrac{3}{2}(1 - U/(3c_0))U^2 = (1 + U/c_0)\Delta p/\rho_0 + R\dot{\Delta p}/(\rho_0 c_0). \tag{2}$$

Its denominator is $R(1 - U/c_0) + 4\mu/(\rho_0 c_0)$, including the viscous acceleration derivative. The original incident-field convention [1] is distinguished from this explicit local forcing convention.

For Gilmore, I use the barotropic Tait law

$$\rho(p) = \rho_0[(p + B)/(p_0 + B)]^{1/n}, \qquad C_L^2 = n(p_L + B)/\rho_L, \tag{3}$$

$$H = \frac{n}{n-1}[(p_L + B)/\rho_L - (p_\infty + B)/\rho_\infty]. \tag{4}$$

The water exponent $n = 7.15$ follows Ref. [2]. I derive $B = \rho_0 c_0^2/n - p_0 = 306.524$ MPa to match the previous ambient acoustic state exactly. Denner's reference uses 304.6 MPa and a slightly different density; those different reference constants are not silently combined. No collapse or temperature is used to calibrate B. The EOS is restricted to $p + B > 0$ and used only for a mechanical sensitivity calculation.

2.2 Explicit acceleration and checks

Gilmore's equation is

$$(1 - U/C_L)R\dot{U} + \tfrac{3}{2}(1 - U/(3C_L))U^2 = (1 + U/C_L)H + (1 - U/C_L)R\dot{H}/C_L. \tag{5}$$

Since $\dot{H} = \dot{p}_L/\rho_L - \dot{p}_\infty/\rho_\infty$, write

$$\dot{p}_L = b_w - 4\mu\dot{U}/R, \qquad b_w = -3\kappa p_d U/R + 2\sigma U/R^2 + 4\mu U^2/R^2, \tag{6}$$

$$D_G = (1 - U/C_L)[R + 4\mu/(C_L\rho_L)], \tag{7}$$

$$N_G = (1 + U/C_L)H - \tfrac{3}{2}(1 - U/(3C_L))U^2 + (1 - U/C_L)\frac{R}{C_L}\left(\frac{b_w}{\rho_L} - \frac{\dot{p}_\infty}{\rho_\infty}\right). \tag{8}$$

Then $\dot{U} = N_G/D_G$, with no clipping. D_G has units m and N_G m^2/s^2. Stable logarithmic/exponential differences evaluate H near equilibrium. Tests compare H with $\int_{p_\infty}^{p_L} \mathrm{d}p/\rho(p)$ and verify its derivative, the full implicit residual, zero-drive equilibrium and the RP limit as the reference sound speed increases.

[2] Mini-Research 2, "Compressible Bubble Dynamics and Extreme Collapse Conditions," Section "Keller–Miksis convention and explicit acceleration," `main.pdf`.

2.3 Results and numerical controls

Table 1: One-cycle comparison, common polytropic gas. $M_0 = \max|U|/c_0$ and $\chi = \max|p_L - p_0|/(\rho_0 c_0^2)$. No gas temperatures are calculated.

P_a (bar)	Model	$R_{\min}$ (μm)	t_R (μs)	$\lvert U_-\rvert$ (m/s)	M_0	χ
1.00	KM	4.393090	20.092624	6.565	0.0044	0.0001
1.00	Gilmore	4.393164	20.092625	6.564	0.0044	0.0001
1.15	KM	1.228160	15.664319	154.233	0.1041	0.0439
1.15	Gilmore	1.233152	15.664393	152.478	0.1029	0.0431
1.30	KM	0.550809	18.341064	1072.981	0.7240	1.2758
1.30	Gilmore	0.513361	18.341191	1031.292	0.6959	1.7149

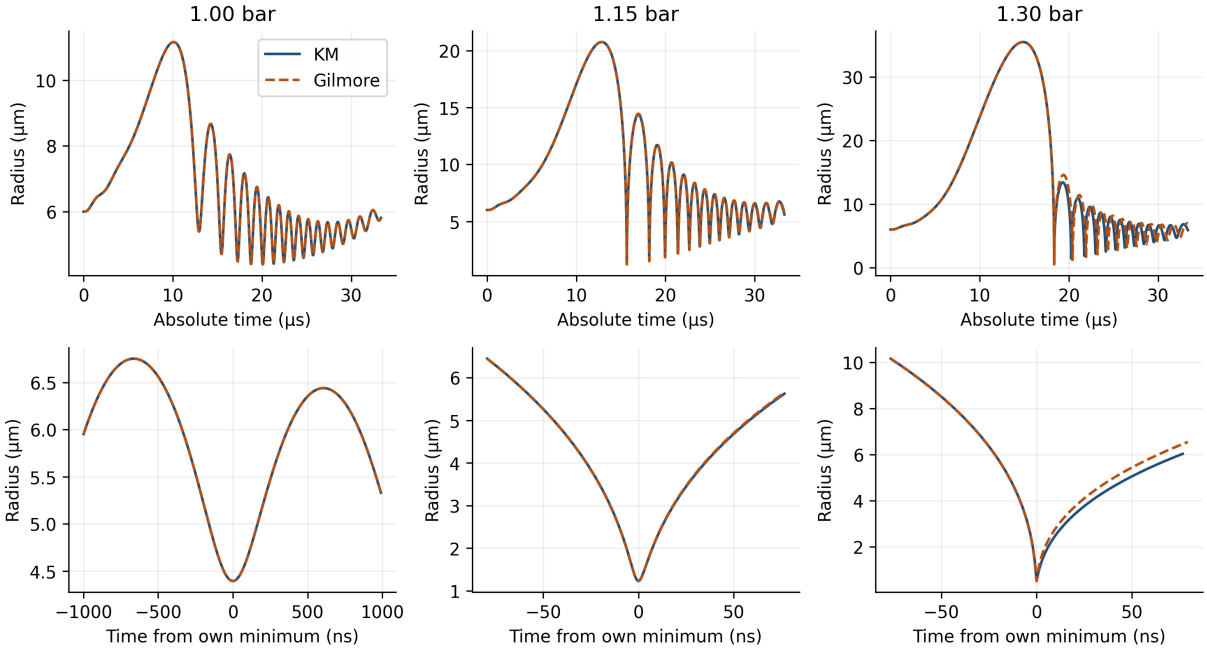


Figure 1: Common absolute-time histories and separately centered minima. At strong collapse, a change of liquid model changes the compression path even with identical gas and forcing.

At 1.15 bar Gilmore raises $R_{\min}$ by about 0.406%; at 1.30 bar it lowers $R_{\min}$ by about 6.80%. The latter has peak inward speed 1031.29 m/s versus 1072.98 m/s for KM. Gilmore's peak wall pressure is 3.76 GPa, maximum Tait density ratio 1.436 and maximum local Mach 0.504. A smaller local Mach does not establish full physical accuracy. A new gas-energy solve would be necessary to determine its temperature; temperatures from MR7 are not mapped onto this polytropic trajectory.

DOP853 uses production relative tolerance 10^{-9} and 800 maximum steps per period, followed by 10^{-11} and 1600. Independent Radau uses 2×10^{-10} and 1200. Component absolute tolerances are respectively 10^{-6}rtol m and 10^{2}rtol m/s. Radius and speed extrema are refined on accepted-step dense histories; endpoints are retained. Across all three amplitudes/models, maximum discrepancies relative to strict DOP853 are 4.27×10^{-10} in $R_{\min}$, 3.70×10^{-9} in peak speed and 1.47×10^{-16} s in the selected event time. These are empirical solver differences, not physical timing accuracy.

At 1.00 bar the shared KM minimum reproduces the earlier first-cycle event. The 1.30 bar first-cycle KM minimum agrees with the saved 0.550808707 μm reference. Both are event correspondence checks, rather than comparisons of unrelated five-cycle global times.

3 What the mechanics cannot establish

First-order KM assumes weak compressibility. In the 1.30 bar common polytrope, $M_0 \simeq 0.724$ and $\chi \simeq 1.276$ are not small parameters. Gilmore introduces a variable-density liquid, but still imposes spherical flow and a barotropic EOS. Neither calculation resolves shock thickness, gas shock focusing, nonspherical jets or a perturbation spectrum. The liquid-model spread is a sensitivity result, not a measurement or an error bar around a known physical solution. Published spherical-model/DNS comparisons have their own tested domains [3].

The minimum far-field pressures are 1325, −13675 and −28675 Pa at 1.00, 1.15 and 1.30 bar. Negative liquid pressure is admissible as a metastable tensile state, while its survival depends on nuclei, contamination, boundaries and duration. The mathematical Tait branch remaining positive does not establish tensile stability. New nuclei and neighboring cavities can destroy the isolated spherical geometry. Conversely, tension alone is not proof that the specified history is physically impossible.

In MR7 the wet 1.15 bar trajectory reaches about 130 MPa gas pressure and 160 kg/m^3 ideal-gas density; wet 1.30 bar reaches about 2.0 GPa and 1157 kg/m^3 before censoring. These outputs challenge the uniform ideal-gas and incompressible thermal-water descriptions. A gas sound-crossing estimate shorter than a thermal width is only one screen; it cannot demonstrate chemical, vibrational or plasma equilibrium at those states.

4 High-temperature thermodynamics

For the later fixed mixture, $p_g = nR_uT/V$ and

$$\Delta U_g = \sum_i n_i \int_{T_0}^{T} [c_{P,i}(\theta) - R_u]\,\mathrm{d}\theta, \quad C_V = \frac{\mathrm{d}U_g}{\mathrm{d}T}, \quad C_V\dot{T} = -p_g\dot{V} + Q_{\ell\to g}. \tag{9}$$

NASA9 heat capacity and consistent integration supply molecular internal excitation, but not composition changes.[3] The archived H_2O domain ends at 6000 K, dry N_2/O_2 at 20000 K. No wet trajectory is continued beyond its coefficients. Ground-state Ar has an analytic constant capacity but misses physical excitation/ionization. Coefficient availability and physical validity are separate questions.

Water vapor, temperature-dependent capacity, evaporation/condensation, dissociation, ionization and nonideal EOS are distinct mechanisms. Fixed saturated water moles do not describe evaporation or latent heat. For reacting species $U = \sum_i n_i u_i(T)$, chemistry enters through $\sum_i u_i\dot{n}_i$, with consistent species and pressure derivatives; external mass exchange also carries enthalpy. For an equilibrium nonideal closure, $\dot{T} = [-(p + U_V)\dot{V} + Q]/U_T$ and $\dot{p} = p_T\dot{T} + p_V\dot{V}$ are required.

NIST dissociation/ionization reference demands identify possible competing storage, not realized conversion.[4] The 15.759610 eV isolated-Ar ionization threshold [10] cannot be converted into an ion fraction by dividing by temperature. No equilibrium composition, Saha fraction or corrected chemical temperature is fabricated. Rate information, density, partition data and relaxation-time evidence are absent.

[3] Mini-Research 4, "Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization," Section "Thermodynamically consistent formulation," `04_Temperature_Barrier.pdf`.

[4] Mini-Research 4, "Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization," Section "Dissociation and ionization: quantitative assessment," `04_Temperature_Barrier.pdf`.

5 Laser and particle feasibility

5.1 Absorption, local heating and phase changes

The nominal inherited Mie cross section is $3.43373245 \times 10^{-15}$ m^2, so one 20 nm-radius particle absorbs $E_{\mathrm{abs}} = 34.3373$ fJ at $F = 10$ J/m^2. The 10 ns top hat has intensity 10^9 W/m^2. For a local 100 μm-radius flat beam, $E_{\mathrm{incident}} = FA = 314.159$ nJ. Only the local particle cross section supplies its source; the full beam or a macroscopic layer's absorption is not assigned to this gas.[5]

The Au capacity is 8.352×10^{-17} J/K. The no-loss sensible equivalent of the nominal absorption is a 411 K rise, whereas the resolved isolated 10 ns conduction control gives about 23.19 K. The latter includes heat leaving Au during the pulse. MR7's radial result is about 6.81 K, and matched stationary radial Au rises are only 21–30% of resolved finite-Au results. A spherical source cannot certify a local peak or material stability. The static discrepancy is not a calibrated error factor for the moving model.

The 30 K Au-rise screen used in earlier moderate designs is not a universal physical maximum. Melting, evaporation, reshaping and changing dielectric response can invalidate high-fluence cold-sphere extrapolations. The bulk 1337.33 K Au point and about 87.2 fJ sensible energy to it at this radius are diagnostic references; latent heat, size, coating and environment remain relevant [9]. The 0.1 ps high-fluence formal MR5 continuation reaches a sensible-equivalent rise of order 4.26×10^4 K, but violates electron/lattice equilibration and phase/optical assumptions. It is preserved as a formal continuation, not an attainable metal temperature.

Local water nucleation can create a second cavity and change contact, optics and flow. The roughly 600 J/m^2 threshold measured by Kitz et al. for supported 90 nm-diameter particles with 5 ns pulses [8] is not a calibrated threshold for this 40 nm-diameter Au sphere beside a moving gas surface. Atmospheric boiling is likewise not a universal nanoscale nucleation criterion.

5.2 Retention near the moving interface

A passive center in the inherited incompressible flow has $r_{\mathrm{np}}^3 - R^3 = \text{constant}$ and moves away from the wall during collapse. Prescribed $r_{\mathrm{np}} = R + a + d$ instead requires relative speed $v_{\mathrm{slip}} = U[1 - (R/r_{\mathrm{np}})^2]$.[6]

P_a (bar)	Required slip (m/s)	Particle Reynolds number
1.00	0.42084	0.01677
1.15	24.437	0.97377
1.30	220.64	8.7922

These new saved-trace screens use $\mathrm{Re}_{\mathrm{np}} = 2a\rho_0|v_{\mathrm{slip}}|/\mu$. The 1.714 ns Stokes relaxation reference does not justify a low-Re force estimate when the formal 1.30 bar case reaches $\mathrm{Re} \simeq 8.79$. No trapping/adhesion/acoustic-force model is supplied. A fixed center can contact the expanding bubble; no source is physically continued into gas after contact. Passive placement, fixed centers and prescribed confinement cannot be interchanged when interpreting the laser effect.

[5] Mini-Research 5, "Plasmonic Absorption and Photothermal Heating of Gold Nanoparticles," Section "Optical theorem, Mie theory and spectral results," `05_Plasmonic_Heating.pdf`.

[6] Mini-Research 6, "Frozen Geometry AuNP–Bubble Thermal Coupling," Section "Kinematics, event resolution and applicability," `06_AuNP_Bubble_Coupling.pdf`.

6 Timing and ideal favorable transfer

The characteristic times at 20 nm radius are approximately 1.7 ps for electron–phonon relaxation, 3.15 ps for internal metal diffusion and 0.166 ns for the Au/water interface storage scale. The pulse lasts 10 ns. Water diffusion over a 100 nm surface gap takes $d^2/\alpha_\ell = 69.8$ ns; a 120 nm center distance gives 100.5 ns, and 500 nm gives 1.745 μs. Diffusion scales are not hard arrival boundaries or proof that useful energy reaches gas before its peak.

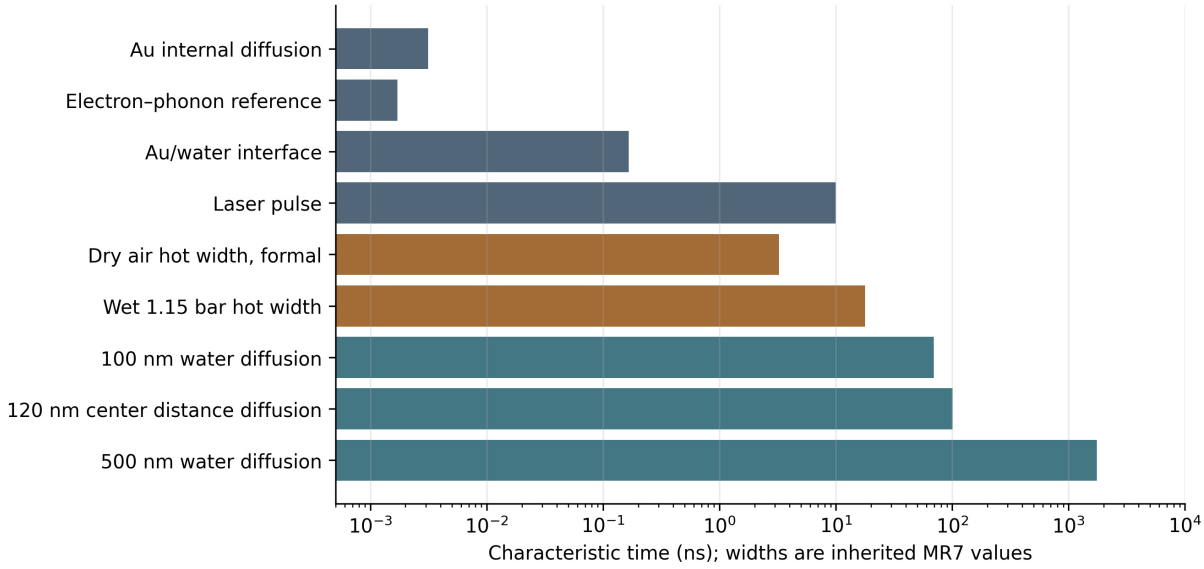


Figure 2: Transport scales and inherited MR7 thermal half-height widths. The water-vapor censored state has no complete width. Widths are not chemical equilibration times or ion-confinement measurements.

The stationary MR6 reference delivers only 0.3258% of its absorption to the gas by 2 μs; it does not measure collapse efficiency. In MR7's early nominal pairs, the net wall-power increment peaks 38.6 ns and 34.9 ns after pulse center at 1.00 and 1.15 bar. Those increments include feedback. At 1.15 bar the early response is comparatively stable, but the late event-centered sensitivity changes by 89.6% under 48-to-96-cell refinement and remains unresolved.

To separate available energy from its modeled delivery, I define an ideal instantaneous fixed-volume operation at each specified baseline peak:

$$\int_{T_b}^{T_*} C_V(T)\,\mathrm{d}T = E_{\mathrm{abs}}. \tag{10}$$

The same coefficient domains and gas moles are retained. This is a favorable direct caloric allowance for that defined operation; it omits transport, losses and later mechanical work. It is neither an achieved temperature increment nor a universal upper bound for a dynamic or local hot region.

Reference T (K)	Frozen gas	Ideal ΔT (K)
553.02	air wet	0.0334742
2798.28	air wet	0.0249898
6263.58	air dry	0.0234249
22808.72	argon dry	0.0590651

The table uses $F = 10$ J/m^2 and one particle. Additional 1, 100 and 1000 J/m^2 algebraic scenarios are archived with their formal status. Wet 6000 K is not continued. The dry-Ar row remains a ground-state frozen-gas idealization. Dividing energy by one constant heat capacity would miss the variable caloric response and would still not provide a universal temperature bound.

7 Signed energy audit and mechanical work

The closed gas satisfies $\Delta U_g + W_{\mathrm{by}} - Q_{\ell g} = 0$. The liquid and Au identities are

$$\Delta E_\ell - Q_{\mathrm{Au}\ell} + Q_{\ell g} + Q_{\mathrm{bath}} = 0, \tag{11}$$

$$\Delta E_{\mathrm{Au}} - E_{\mathrm{abs}} + Q_{\mathrm{Au}\ell} = 0, \tag{12}$$

$$\Delta(U_g + E_\ell + E_{\mathrm{Au}}) + W_{\mathrm{by}} + Q_{\mathrm{bath}} - E_{\mathrm{abs}} = 0. \tag{13}$$

Positive $Q_{\ell g}$ heats gas and negative values cool it. Compression has negative W_{by}. Each internal flux cancels in the sum. These are thermal-subsystem identities, excluding liquid kinetic energy, acoustic drive energy, viscous heat and possible retaining-force work. With mass exchange or chemistry, the balances would require the corresponding additional channels.

Table 2: Saved endpoint energies, reassembled in MR8, in pJ. The first two endpoints are five-cycle endpoints; 1.30* stops at its observed coefficient boundary. Acoustic background is included, so these are not laser-tagged storage amounts.

P_a (bar)	E_{abs}	E_{Au}	E_ℓ	ΔU_g	W_{by}	$Q_{\ell g}$
1.00	0.0343373	4.64855e-06	1157.09	-7.67078	-1148.83	-1156.5
1.15	0.0343373	1.4756e-05	3831.97	0.448837	-3842.28	-3841.83
1.30*	0.0343373	9.83682e-05	-210.908	7664.12	-7452.98	211.147

The identical absorbed energy in each row is only 0.0343373 pJ from the local 314.159 nJ beam. Water can have signed negative storage relative to the ambient reference without violating conservation. Net gas heat can also be negative while compression supplies positive sensible energy. The actual signs are preserved.

Recomputing the four identities from independently saved quantities gives maximum total residual 1.35×10^{-6} of the run thermal scale. For nonlinear laser increments I independently evaluate

$$\delta U_g = \delta T \int_0^1 C_V(T_b + s\delta T)\,\mathrm{d}s, \tag{14}$$

using the saved common-prefix temperatures, and compare it with $-\delta W_{\mathrm{by}} + \delta Q_{\ell g}$. The maximum paired gas residual is 4.61×10^{-8} of absorption. Existing paired total residuals, derived from the original water/Au states, remain below 4.72×10^{-8} of absorption in the selected traces. Those original full paired water states are not reconstructed by this audit; their independently saved residuals and ledgers are retained.

At the complete 1.00 bar endpoint, the new caloric gas increment is 4.50×10^{-19} J, while work-by-gas and gas-heat increments are approximately -7.79×10^{-18} and -7.34×10^{-18} J. The nominal global peak difference is positive. Different signs at an endpoint and at a peak are not contradictory. Changes in compression work prevent interpreting a gas-energy difference or net wall-heat difference as a pure optical efficiency.

Numerical energy consistency does not validate high-temperature chemistry, local material safety or the unmodeled mechanical energy budget. The new twelve-check suite separates mathematical identities, numerical convergence, domain enforcement and preservation of previous inputs.

8 Extreme scenarios, thermonuclear feasibility and conclusions

Representative scenario	Temperature or effect	Interpretation
Wet air, MR7, 1.00 bar, nominal pulse	553.023 K baseline; $+1.408816\,\mu$K	Complete mathematical pair; conditional reduced geometry/transport and unverified retention
Wet air, MR7, 1.15 bar, nominal pulse	2798.281 K baseline; $+2.184879\,\mu$K	Complete numerical pair, physically formal composition/EOS/flow; late timing branch unresolved
Wet air, MR7, 1.30 bar	Observed 6000 K boundary; global effect undefined	Censored and formal; coefficient cutoff is not a ceiling
Dry MR7, 1.30 bar	6263.58 K air; 22808.7 K ground-state Ar	Complete formal frozen-gas extrema; high-density/reacting thermodynamics absent
MR3 adiabatic RP, 1.295 bar	1.13046×10^6 K complete interval	Formal incompressible extrapolation; invalid wall-speed and gas closure
MR3 adiabatic RP, 1.185 bar	Observed 1.16720×10^6 K	Technical-boundary censored; global maximum unknown
Rare spatial hot region with fuel	$T_i, T_e, n_D, V_{\mathrm{hot}}, \tau$ unknown	Unresolved physical scenario; no probability or yield assigned

The MR3 formal extrema are preserved in their original scope.[7] The new Gilmore result supplies mechanical sensitivity only; it neither validates the RP values nor furnishes a new thermodynamic maximum. Numerically formal extrema are not experimentally confirmed states. Failure of a reduced model to resolve a rare configuration is not proof of its impossibility.

For identical deuterons, a physically justified local reaction calculation would require

$$N_{\mathrm{DD}} = \int \tfrac{1}{2} n_D^2 \langle \sigma v \rangle_{\mathrm{DD}} \, \mathrm{d}V \, \mathrm{d}t. \tag{15}$$

The uniform scalar gas temperature is not an ion temperature, electron temperature or spatial hot-region maximum. This program has no deuterium-enriched fuel specification, isotope-resolved reacting density, velocity distribution, hot volume or reacting lifetime. Standard nuclear reactivity data exist [14], but their prerequisites are absent. The reaction count is left undetermined, not assigned zero or evaluated with arbitrary inputs.

The deuterated-acetone calculations of Ref. [11] use different temperatures, acoustic conditions, radii, EOS and laser coupling; their Table III and Table IV are not a matched AuNP pair. Their hotter-core discussion is an interpretation, not a measurement of this water system. Acid-host spectroscopy [12, 13] likewise distinguishes observed emission temperatures from possible core conditions and supplies no direct calibration of T_i here.

The nominal single-particle channel changes the computed peak very little, but no universal laser-efficiency limit or physically reliable maximum temperature has been established. The present program does not substantiate thermonuclear feasibility, including in rare extreme configurations; the physically admissible extreme state remains unresolved. The minimal discriminating additions are a measured drive/radius/composition history, coupled compressible and reacting nonideal thermodynamics, finite-Au moving local transport with realizable retention and phase-dependent optics, and, for a nuclear assessment, resolved fuel and ion density–volume–time data with independent diagnostics. A full new CFD or fusion simulator has not been substituted for these missing premises.

[7] Mini-Research 3, "In-Bubble Thermodynamics: Compression Heating, Energy Balance, and Extreme Temperature Estimates," Section "Applicability of extreme scenarios," `03_In_Bubble_Thermodynamics.pdf`.

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